In-Situ Time Resolved Powder Diffaction Studies in Heterogenous Catalysis; Coupling the Study of Long Range and Local Structural Changes

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Introduction

The application of in-situ time resolved powder diffraction to study structural changes that occur during chemical reactions has become a standard method at synchrotron sources. With respect to heterogeneous catalysis, the understanding of the structural changes, both on the local and long range length scales, that are the consequence of reaction conditions, is paramount to understanding fully the functioning of many catalysts. Time resolved diffraction experiments which *combine* high-energy X-rays (>80 keV) with area detectors open the possibility of coupling both reciprocal and real space methods to probe both long range and local structural changes simultaneously [1]. This is particularly advantageous in the study of chemically induced structural changes, commonly encountered in heterogeneous catalysis. Here we illustrate the application of the approach to monitor the reduction of CuO, a material that is ubiquitous in compositions of methanol synthesis and water-gas shift catalysts [2]. Activation of the catalysts requires reduction of the CuO, producing small metallic Cu particles which are the active centers for catalysis. We have therefore been studying the reduction of CuO under controlled conditions using H_2 to examine the formation of centers active for catalysis (3).

Experimental

All diffraction experiments were performed at the 1-ID beamline at the Advanced Photon Source (APS) located at Argonne National Laboratory, Argonne, IL (USA). High energy X-rays were selected using a double bent Laue monochrometer capable of providing a flux of 10^{12} photons/second and operating with X-rays in the energy range of 50-200 keV.

In-situ diffraction experiments were performed with a specially designed "flow cell" (Figure 1), which allows the flow of reactant gas over the sample during acquisition of powder diffraction data. The temperature was controlled in steps as follows; 2 hour ramp from room temperature to 350ºC. During the experiment 4%H_2 in He was passed over the sample at a rate of 8 cm^3 per minute.

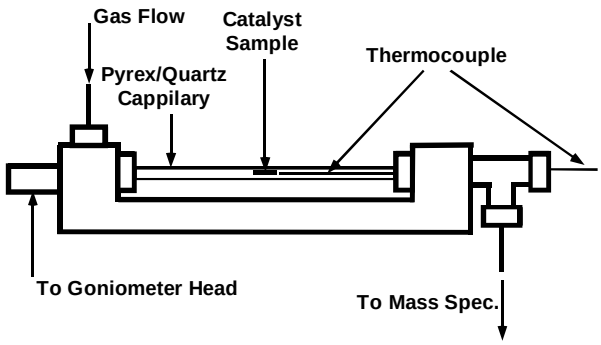


Figure 1. *Diagram of flow cell for in-situ experiments.*

A MAR345 image plate camera was mounted orthogonal to the beam path, with the beam centered on the IP. When using the 99 keV X-rays, a LaB_6 standard was used to calibrate the sample to detector distance and tilt of the IP relative to the beam path, using the software Fit2D [4]. All raw two-dimensional image data were integrated using the software Fit2D and files output as intensity versus 2θ. Ten second exposures yielded data that provided very good statistics for the refinement techniques described below.

Results

The in-situ reduction of CuO was followed using high energy X-rays coupled with a two dimensional image plate (IP) detector. The results are shown in Figure 2.

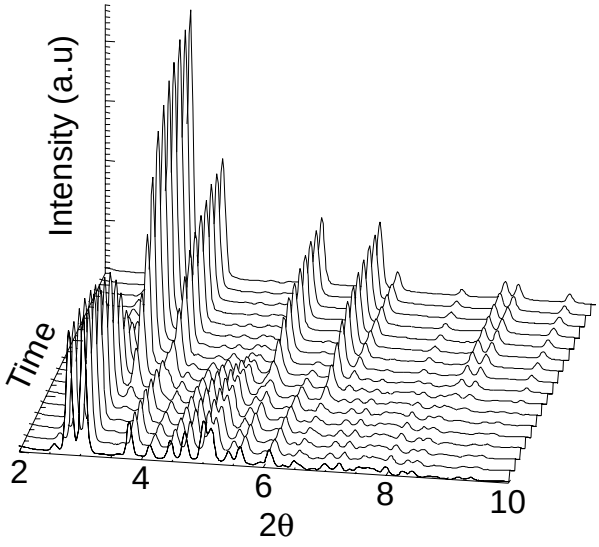


Figure 2. *Time resolved in-situ powder diffraction of the reduction of CuO.*

The reduction of CuO to Cu metal takes place at approximately 280ºC. The reduction to Cu is quite fast and occurs on the order of minutes. During this period both CuO and Cu are evident, and we are currently investigating the structural **changes** that occur.

High energy X-rays provide several experimental advantages in the structural analysis of time resolved data. Utilization of high-energy X-rays allow for larger values of momentum transfer to be probed (> 18 $Å^{-1}$), opening up the possibility for the application of real space methods (pair distribution function, PDF, analysis) to probe changes to local structure during reaction conditions. The G(r)s obtained from the diffraction patterns (IP data) at the beginning of the in-situ experiment and at the end of the in-situ experiment are shown in Figure 3. For example, the first Cu-O peak drops dramatically and a new peak at 2.6 Å (Cu-Cu) is observed. Traditional Bragg refinement of the same diffraction data used for PDF analysis provides for an improved parameter to data ratio. Exceptional Bragg profile refinements of beginning and final phases can be obtained, but disorder in mixed phase results in poorer fits and significant features in the difference electron density maps, which we are attempting to interpret. Following the reduction using real space methods may open up the further understanding of the reduction process, as the changes to the Cu-O coordination during the final stages of reduction may not be evident in the Bragg profile refinements.

Conclusions

High energy X-rays open up the possibility for qualitatively new time-resolved experiments which couple both reciprocal and real space approaches to structural analysis. The application of X-rays with energies between 80-120 keV allows for collection of powder diffraction data to high momentum transfer (>18 $Å^{-1}$) using commercial image plate cameras at sampling rates which will allow monitoring relevant structure changes under process conditions. This allows for both Rietveld and PDF analysis to be carried out on the same data sets probing both long range and local structural changes simultaneously.

Acknowledgements

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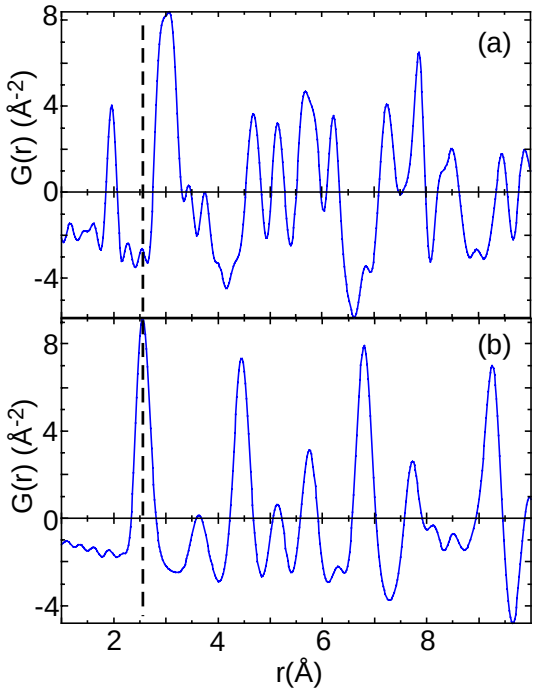


Figure 3. *G(r) for (a) CuO at the beginning of the time resolved experiment and G(r) for (b) Cu formed at the end of the time resolved experiment. The guideline shows the Cu-Cu first shell distance.*

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